RAFFLES INSTITUTION
H2 Mathematics (9758)
2025 Year 5

Chapter 8B: Curve Transformation

SYLLABUS INCLUDES

- Effect of transformations on the graph of $y = \mathrm{f}(x)$ as represented by $y = a\mathrm{f}(x)$, $y = \mathrm{f}(x) + a$, $y = \mathrm{f}(x + a)$ and $y = \mathrm{f}(ax)$, and combinations of these transformations
- Relating the graphs of $y = |\mathrm{f}(x)|$, $y = \mathrm{f}(|x|)$, and $y = \dfrac{1}{\mathrm{f}(x)}$ to the graph of $y = \mathrm{f}(x)$
- Relating the graph of $y = \mathrm{f}'(x)$ to the graph of $y = \mathrm{f}(x)$

CONTENT

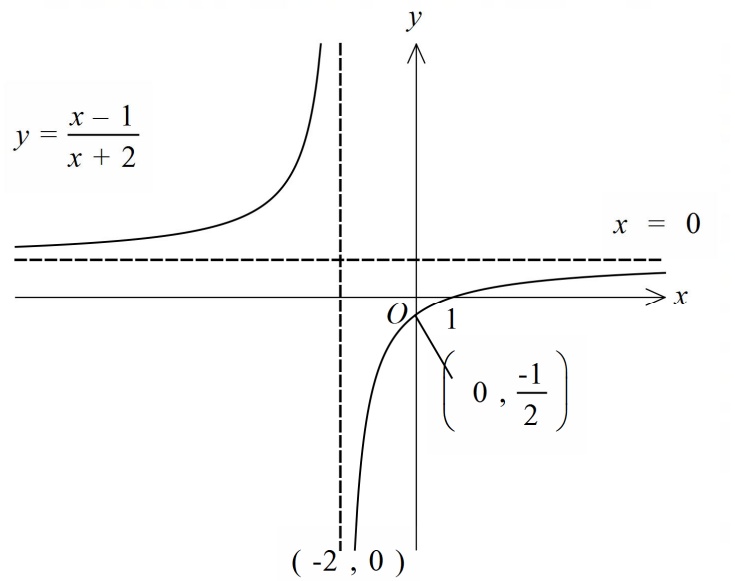
1 Standard Transformations
- 1.1 Translation
- 1.2 Scaling (Stretching)
- 1.3 Reflection about axes
- 1.4 Combination of standard transformations

2 More Transformations
- 2.1 Graphs of modulus functions
- 2.2 Graphs of derivative functions
- 2.3 Sketching the graph of $y = \dfrac{1}{\mathrm{f}(x)}$ from the graph of $y = \mathrm{f}(x)$

In mathematics, the study of functions and their graphical representations provides a powerful tool for understanding the dynamic relationships between variables. Transformations of graphs play a pivotal role in this exploration, allowing mathematicians to analyze how a function evolves and adapts when subjected to various modifications. These transformations encompass a range of operations such as translations, reflections, scaling, each imparting distinct changes to the original graph. Through the lens of transformations, mathematicians gain insights into the behaviour of functions and unravel the intricate patterns that underlie their geometric alterations.

1 STANDARD TRANSFORMATIONS

Consider the curve given by $y=\dfrac{x-1}{x+2}$.



We notice that the shape of the graph is similar to the graph of the reciprocal function $\mathrm{f}(x)=\dfrac{1}{x}$.

In fact, we can write $y=\dfrac{x-1}{x+2}=1-\dfrac{3}{x+2}=1-3\,\mathrm{f}\left(x+2\right)$.

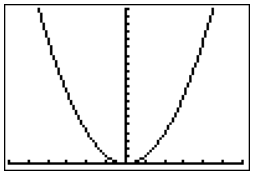
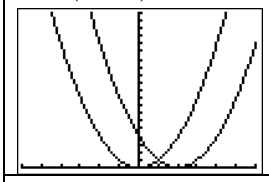
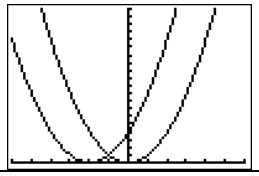
From the form $y=1-3\,\mathrm{f}\left(x+2\right)$, we observe that there is a sequence of transformations that would transform the curve $y=\dfrac{1}{x}$ onto the curve $y=\dfrac{x-1}{x+2}$. We will learn more about these in the following sections.

1.1 Translation

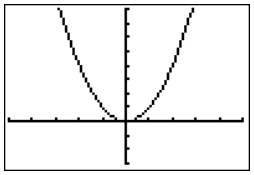
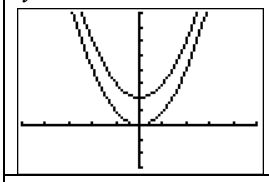
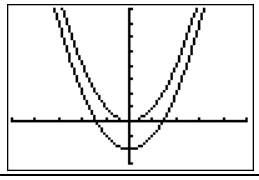
Translating a graph in the direction of an axis is to move the graph in the direction of the axis, **without** changing its *shape* or *size.*

Consider $\mathrm{f}(x) = x^2$.

(A) Compare the graphs of $y = \mathrm{f}(x) = x^2$, $y = \mathrm{f}(x-2) = (x-2)^2$ and $y = \mathrm{f}(x+2) = (x+2)^2$.

$y = \mathrm{f}(x)$	Transformed equation $y = \mathrm{f}(x \pm a),\ a > 0$	Replacement	Description of transformation	Remarks
$y = x^2$ 	$y = (x-2)^2$ 	Replace x by $x-2$	**Translation** of the graph $y = x^2$ by 2 units in the **positive** x-direction.	NO change in y values
	$y = (x+2)^2$ 	Replace x by $x+2$	**Translation** of the graph $y = x^2$ by 2 units in the **negative** x-direction.	

(B) Compare the graphs of $y = \mathrm{f}(x) = x^2$, $y = \mathrm{f}(x) + 2 = x^2 + 2$ and $y = \mathrm{f}(x) - 2 = x^2 - 2$.

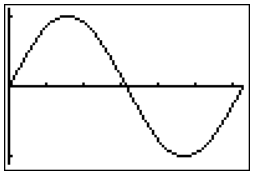
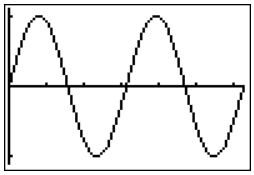
$y = \mathrm{f}(x)$	Transformed equation $y = \mathrm{f}(x) \pm a,\ a > 0$	Replacement	Description of transformation	Remarks
$y = x^2$ 	$y = x^2 + 2$ 	Replace y by $y-2$	**Translation** of the graph $y = x^2$ by 2 units in the **positive** y- direction.	NO change in x values
	$y = x^2 - 2$ 	Replace y by $y+2$	**Translation** of the graph $y = x^2$ by 2 units in the **negative** y- direction.	

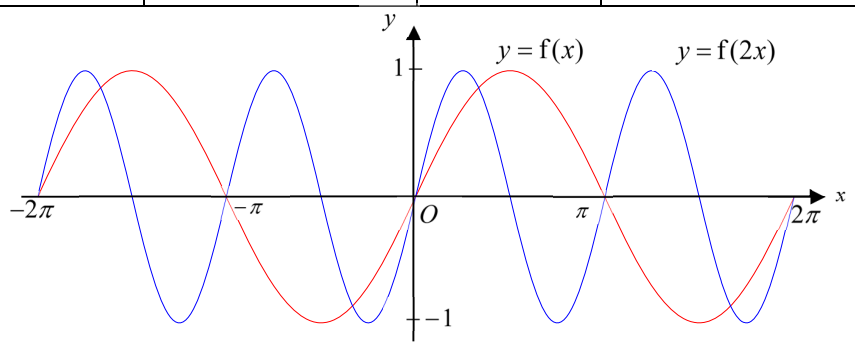
1.2 Scaling (Stretching)

Scaling/stretching a curve parallel to one axis (with the other axis being invariant) changes the *size* of the curve.

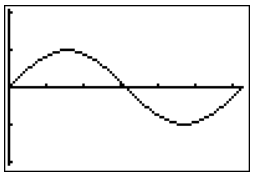
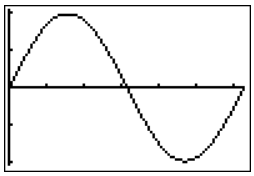
Consider $\mathrm{f}(x) = \sin x$, $0 \le x \le 2\pi$.

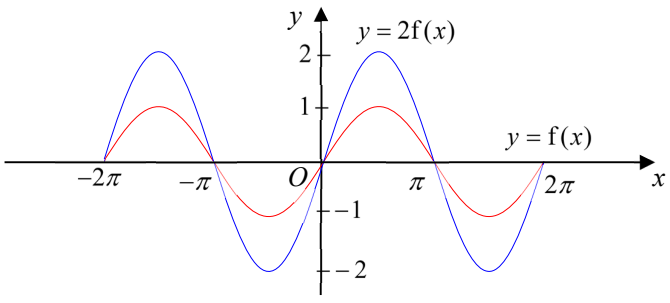
(C) Compare the graphs of $y = \mathrm{f}(x) = \sin x$ and $y = \mathrm{f}(2x) = \sin 2x$.

$y = \mathrm{f}(x)$	Transformed equation $y = \mathrm{f}(ax)$, $a > 0$	Replacement	Description of transformation	Remarks
$y = \sin x$ 	$y = \sin 2x$ 	Replace x by $2x = \dfrac{x}{\left(\frac{1}{2}\right)}$	**Scaling** parallel to the ***x*-axis** of the graph of $y = \sin x$ by a **factor** of $\frac{1}{2}$.	NO change in y values



(D) Compare the graphs of $y = \mathrm{f}(x)$ and $y = 2\mathrm{f}(x) = 2\sin x$.

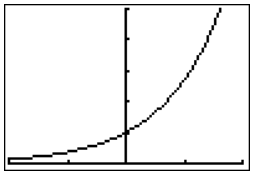
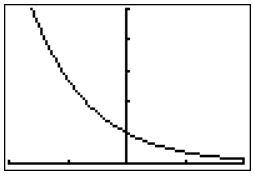
$y = \mathrm{f}(x)$	Transformed equation $y = a\mathrm{f}(x)$, $a > 0$	Replacement	Description of transformation	Remarks
$y = \sin x$ 	$y = 2\sin x$ 	Replace y by $\dfrac{y}{2}$	**Scaling** parallel to the ***y*-axis** of the graph of $y = \sin x$ by a **factor** of 2.	NO change in x values

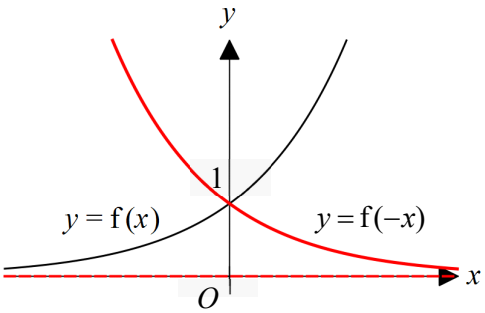


1.3 Reflection about axes

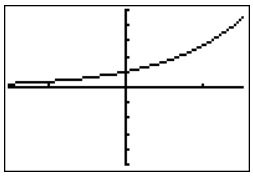
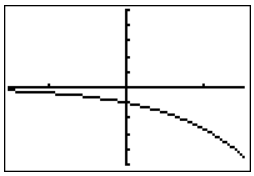
Consider $\mathrm{f}(x)=\mathrm{e}^{x}$.

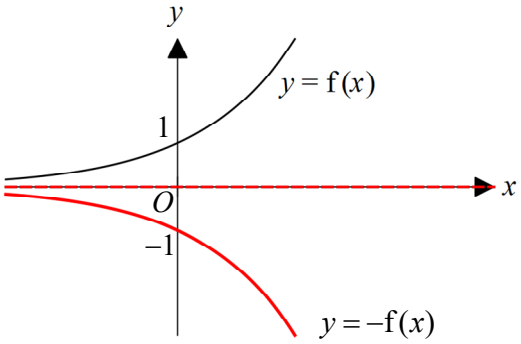
(E) Compare the graphs of $y=\mathrm{f}(x)=\mathrm{e}^{x}$ and $y=\mathrm{f}(-x)=\mathrm{e}^{-x}$.

$y=\mathrm{f}(x)$	Transformed equation $y=\mathrm{f}(-x)$	Replacement	Description of transformation	Remarks
$y=\mathrm{e}^{x}$ 	$y=\mathrm{e}^{-x}$ 	Replace x by $-x$	**Reflection** of the graph of $y=\mathrm{e}^{x}$ in the ***y*-axis**.	NO change in y values



(F) Compare the graphs of $y=\mathrm{f}(x)=\mathrm{e}^{x}$ and $y=-\mathrm{f}(x)=-\mathrm{e}^{x}$.

$y=\mathrm{f}(x)$	Transformed equation $y=-\mathrm{f}(x)$	Replacement	Description of transformation	Remarks
$y=\mathrm{e}^{x}$ 	$y=-\mathrm{e}^{x}$ 	Replace y by $-y$	**Reflection** of the graph of $y=\mathrm{e}^{x}$ in the ***x*-axis**.	NO change in x values

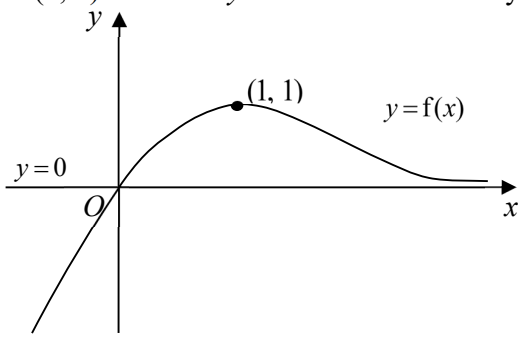


In general, the effects of transformations on the graph of $y = \mathrm{f}(x)$ are summarised in the table below. **Note that *a* is taken to be a positive constant.**

Replacement of x / y	**Transformed Equation**	**Description of transformation**
Replace x by $x-a$	$y = \mathrm{f}(x-a)$	Translation of a units in the positive x-direction
Replace x by $x+a$	$y = \mathrm{f}(x+a)$	Translation of a units in the negative x-direction
Replace y by $y-a$	$y-a = \mathrm{f}(x)$ i.e. $y = \mathrm{f}(x)+a$	Translation of a units in the positive y-direction
Replace y by $y+a$	$y+a = \mathrm{f}(x)$ i.e. $y = \mathrm{f}(x)-a$	Translation of a units in the negative y-direction
Replace x by $\frac{x}{a}$	$y = \mathrm{f}\left(\frac{x}{a}\right)$	Scaling parallel to the x-axis by factor of a (in terms of coordinates, no change in corresponding y-coordinate)
Replace y by $\frac{y}{a}$	$\frac{y}{a} = \mathrm{f}(x)$ i.e. $y = a\mathrm{f}(x)$	Scaling parallel to the y-axis by factor of a (in terms of coordinates, no change in corresponding x-coordinate)
Replace x by $-x$	$y = \mathrm{f}(-x)$	Reflection in the y-axis
Replace y by $-y$	$-y = \mathrm{f}(x)$ i.e. $y = -\mathrm{f}(x)$	Reflection in the x-axis

Example 1

The diagram below shows the graph of $y = \mathrm{f}(x)$. The curve passes through the origin and has a maximum turning point at (1, 1). The line $y = 0$ is a horizontal asymptote to the curve.



Sketch, on separate diagrams, the graphs of

(a) $y = \mathrm{f}(x+2)$, **(b)** $y = 2\,\mathrm{f}(x)$, **(c)** $y = \mathrm{f}(-x)$,

labelling clearly any intersections with the axes, turning points and asymptotes.

Solution

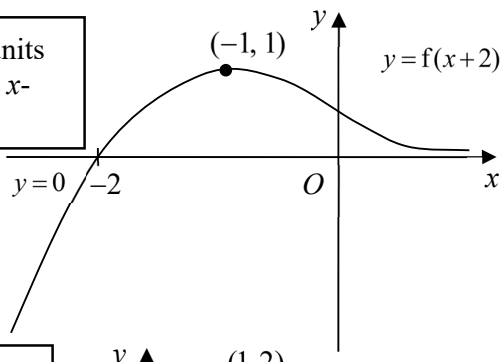
(a)

$y = \mathrm{f}(x)$

↓ Replace x by $x+2$

$y = \mathrm{f}(x+2)$

* Translate 2 units in the negative x-direction



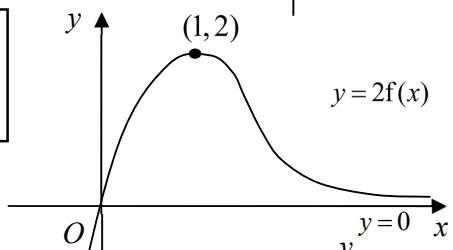
(b)

$y = \mathrm{f}(x)$

↓ Replace y by $\frac{y}{2}$

$y = 2\,\mathrm{f}(x)$

* Scale parallel to the y-axis by a factor of 2



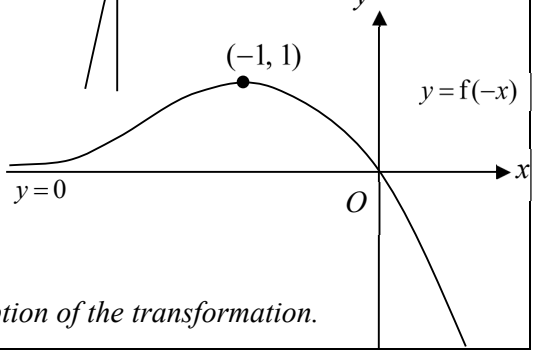
(c)

$y = \mathrm{f}(x)$

↓ Replace x by $-x$

$y = \mathrm{f}(-x)$

* Reflect in the y-axis



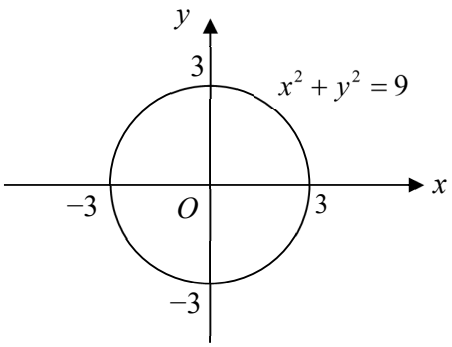
* *Answers required if the question requires a description of the transformation.*

Example 2 (Reference: Chapter 5D, Section 3.2)

Recall that the equation of a circle with centre (0,0) and radius 3 units is given by

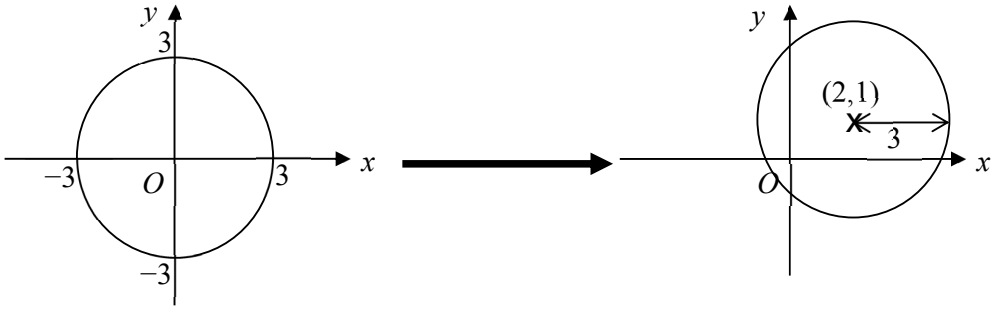
$$x^2 + y^2 = 9$$

.



Describe a sequence of transformations that will transform the above circle to the circle with centre (2,1) and radius 3 units, and state the equation of the new circle.

Solution



$x^2 + y^2 = 9$ $\xrightarrow{\text{Replace } x \text{ by } (x-2)}$ $(x-2)^2 + y^2 = 9$ $\xrightarrow{\text{Replace } y \text{ by } (y-1)}$ $(x-2)^2 + (y-1)^2 = 9$

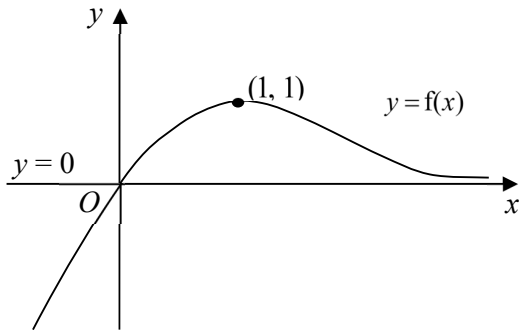
The sequence of transformations required is
1. Translation of 2 units in the positive *x*-direction.
2. Translation of 1 unit in the positive *y*-direction.

Hence, using our knowledge on transformation, the equation of a circle with centre (2,1) and radius 3 units is $(x-2)^2 + (y-1)^2 = 9$.

1.4 Combination of Standard Transformations

In this section, we look at the effects of a series of transformations on a given graph.

Example 3 (same graph as Example 1)
The diagram below shows the graph of $y = \mathrm{f}(x)$. The curve passes through the origin and has a maximum turning point at (1, 1). The line $y = 0$ is a horizontal asymptote to the curve.



Sketch, on separate diagrams, the graphs of

(a) $y = 2\mathrm{f}(x+3)$, (b) $y = \mathrm{f}(-x)+3$, (c) $y = \frac{1}{2}\mathrm{f}\left(\frac{x}{3}\right)$,

(d) $y = \mathrm{f}\left(\frac{x}{3}-1\right)$, (e) $y = \mathrm{f}(3-x)$, (f) $y = 2\,\mathrm{f}(x)+3$,

labelling clearly any intersections with the axes, turning points and asymptotes, whenever possible.

Solution

(a)

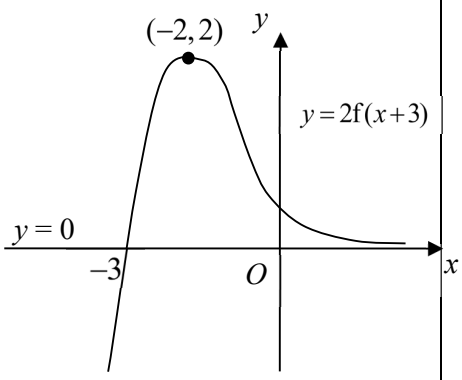
$y = \mathrm{f}(x)$

↓ Replace y by $\frac{y}{2}$ ← * Scale parallel to the y-axis by a factor of 2

$\frac{y}{2} = \mathrm{f}(x) \Leftrightarrow y = 2\,\mathrm{f}(x)$

↓ Replace x by $x+3$ ← * Translate 3 units in the negative x-direction

$y = 2\,\mathrm{f}(x+3)$



Q: Can we choose to translate first then scale?

$y = \mathrm{f}(x)$

↓ Replace x by $x+3$

$y = \mathrm{f}(x+3)$

↓ Replace y by $\frac{y}{2}$

$\frac{y}{2} = \mathrm{f}(x+3) \Leftrightarrow y = 2\,\mathrm{f}(x+3)$

(b)

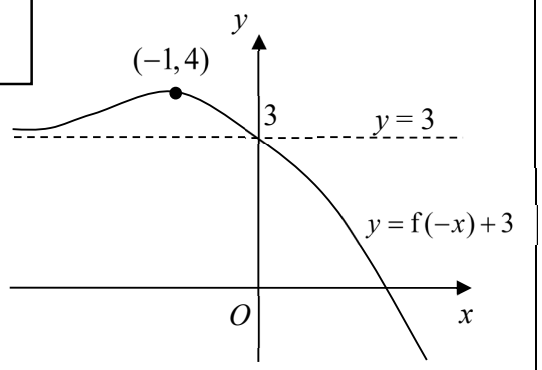
$y = \mathrm{f}(x)$

$\downarrow$ Replace x by $-x$ ⟸ * Reflect in the y-axis

$y = \mathrm{f}(-x)$

$\downarrow$ Replace y by $y - 3$ ⟸ * Translate 3 units in the positive y-direction

$y - 3 = \mathrm{f}(-x) \Leftrightarrow y = \mathrm{f}(-x) + 3$



(c)

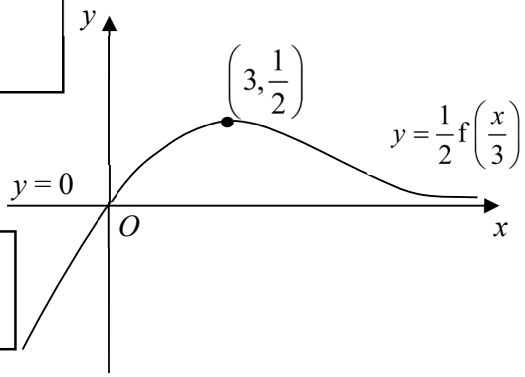
$y = \mathrm{f}(x)$

$\downarrow$ Replace y by $\dfrac{y}{\frac{1}{2}}$ ⟸ * Scale parallel to the y-axis by a factor of $\dfrac{1}{2}$

$\dfrac{y}{\frac{1}{2}} = \mathrm{f}(x) \Leftrightarrow y = \dfrac{1}{2}\mathrm{f}(x)$

$\downarrow$ Replace x by $\dfrac{x}{3}$ ⟸ * Scale parallel to the x-axis by a factor of 3

$y = \dfrac{1}{2}\mathrm{f}\left(\dfrac{x}{3}\right)$



(d)

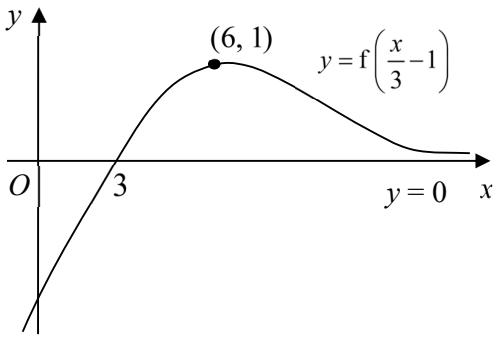
$y = \mathrm{f}(x)$

↓ Replace x by $x-1$ ← * Translate 1 unit in the positive x-direction

$y = \mathrm{f}(x-1)$

↓ Replace x by $\frac{x}{3}$ ← * Scale parallel to the x-axis by a factor of 3

$y = \mathrm{f}\left(\frac{x}{3}-1\right)$



Q: Can we choose to scale first then translate?

$y = \mathrm{f}(x)$

↓ Replace x by $\frac{x}{3}$

$y = \mathrm{f}\left(\frac{x}{3}\right)$

↓ Replace x by $x-3$

$y = \mathrm{f}\left(\frac{x-3}{3}\right) \Leftrightarrow y = \mathrm{f}\left(\frac{x}{3}-1\right)$

(e)

$y = \mathrm{f}(x)$

↓ Replace x by $x+3$ ← * Translate 3 units in the negative x-direction

$y = \mathrm{f}(x+3)$

↓ Replace x by $-x$ ← * Reflect in the y-axis

$y = \mathrm{f}(-x+3)$

Q: Can we choose to reflect first then translate?

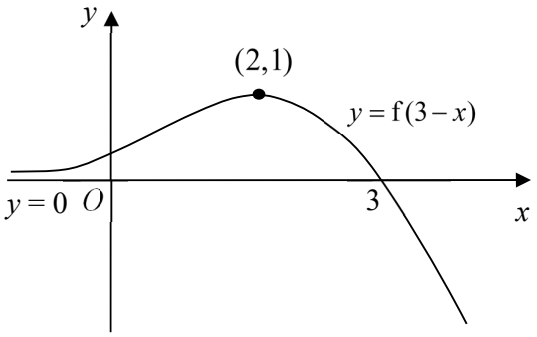
$y = \mathrm{f}(x)$

↓ Replace x by $-x$

$y = \mathrm{f}(-x)$

↓ Replace x by $x-3$

$y = \mathrm{f}\left[-(x-3)\right] \Leftrightarrow y = \mathrm{f}(-x+3)$



(f)

$y = \mathrm{f}(x)$

↓ Replace y by $\frac{y}{2}$ ← * Scale parallel to the y-axis by a factor of 2

$\frac{y}{2} = \mathrm{f}(x) \Leftrightarrow y = 2\,\mathrm{f}(x)$

↓ Replace y by $y - 3$ ← * Translate 3 units in the positive y-direction

$y - 3 = 2\,\mathrm{f}(x) \Leftrightarrow y = 2\,\mathrm{f}(x) + 3$

Q: Can we translate first then scale?

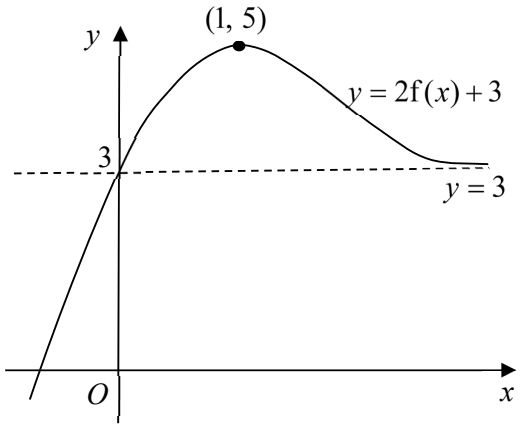
$y = \mathrm{f}(x)$

↓ Replace y by $y - \frac{3}{2}$

$y - \frac{3}{2} = \mathrm{f}(x) \Leftrightarrow y = \mathrm{f}(x) + \frac{3}{2}$

↓ Replace y by $\frac{y}{2}$

$\frac{y}{2} = \mathrm{f}(x) + \frac{3}{2} \Leftrightarrow y = 2\,\mathrm{f}(x) + 3$



Example 4

A curve $y = \mathrm{f}(x)$ undergoes, in succession, the following transformations:

A: A translation of magnitude 2 units in the direction of the *x*-axis
B: A scaling parallel to the *x*-axis by a factor $\frac{1}{2}$
C: A reflection in the *y*-axis
D: A translation of magnitude 1 unit in the direction of the *y*-axis

The resulting curve has equation $y = 1 + \mathrm{e}^{2(x+1)}$. Determine the equation of the curve before the four transformations were effected.

Solution

$y = 1 + \mathrm{e}^{2(x+1)}$

$\downarrow$ before D: Replace y by $y+1$

$y + 1 = 1 + \mathrm{e}^{2(x+1)} \Leftrightarrow y = \mathrm{e}^{2(x+1)}$

$\downarrow$ before C: Replace x by $-x$

$y = \mathrm{e}^{2(-x+1)}$

$\downarrow$ before B: Replace x by $\frac{x}{2}$

$y = \mathrm{e}^{2\left(-\frac{x}{2}+1\right)} \Leftrightarrow y = \mathrm{e}^{-x+2}$

$\downarrow$ before A: Replace x by $x+2$

$y = \mathrm{e}^{-(x+2)+2} \Leftrightarrow y = \mathrm{e}^{-x}$

The equation of the curve is $y = \mathrm{e}^{-x}$.

OR

$y = \mathrm{f}(x)$

$\downarrow A$: Replace x by $x-2$

$y = \mathrm{f}(x-2)$

$\downarrow B$: Replace x by $\frac{x}{\frac{1}{2}}$

$y = \mathrm{f}(2x-2)$

$\downarrow C$: Replace x by $-x$

$y = \mathrm{f}(-2x-2)$

$\downarrow D$: Replace y by $y-1$

$y = \mathrm{f}(-2x-2)+1$

$y = 1 + \mathrm{f}(-2(x+1))$

$\therefore \mathrm{f}(x) = \mathrm{e}^{-x}$

2 MORE TRANSFORMATIONS

2.1 Graphs of modulus functions

The absolute value of a real number x, denoted by $|x|$, is defined by $|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$

Thus we have

$y = \lvert f(x) \rvert = \begin{cases} f(x) & \text{, if } f(x) \geq 0 \\ -f(x) & \text{, if } f(x) < 0 \end{cases}$	$y = f(\lvert x \rvert) = \begin{cases} f(x), & x \geq 0, \\ f(-x), & x < 0. \end{cases}$
The graph of $y = \lvert f(x) \rvert$ lies above (or on) the x-axis.	The graph of $y = f(\lvert x \rvert)$ is symmetrical about the y-axis.
To obtain this graph from the graph of $y = f(x)$:	
1. Retain the part of the graph of $y = f(x)$ where $f(x) \geq 0$. **2. Reflect** in the x – axis the part of the graph of $y = f(x)$ where $f(x) < 0$.	**1. Retain** the part of the graph of $y = f(x)$ where $x \geq 0$ **2. Reflect** the part drawn in **1.** in the y – axis so that the graph of $y = f(\lvert x \rvert)$ is symmetrical about the y-axis.

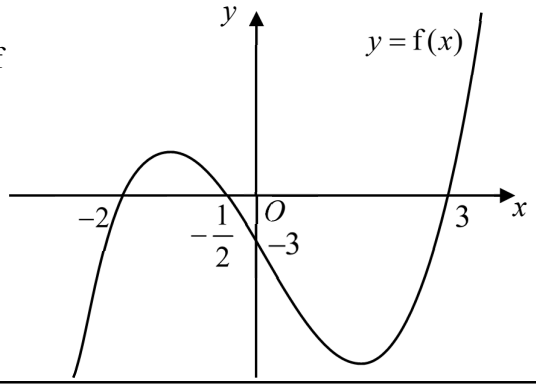
Example 5

The diagram shows the graph of $y = f(x)$.

Sketch, on separate diagrams, the graphs of

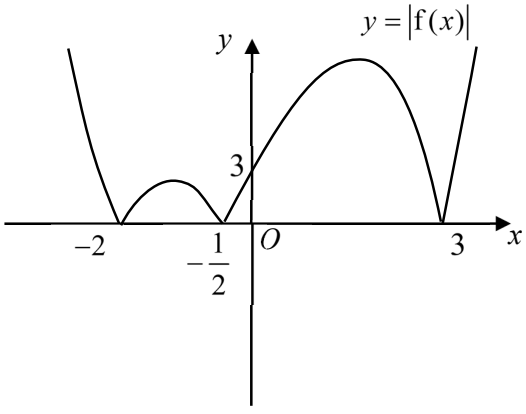
(i) $y = \lvert f(x) \rvert$,

(ii) $y = f(\lvert x \rvert)$.

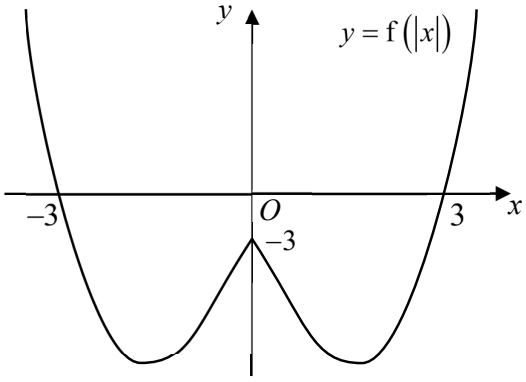


Solution

(i)

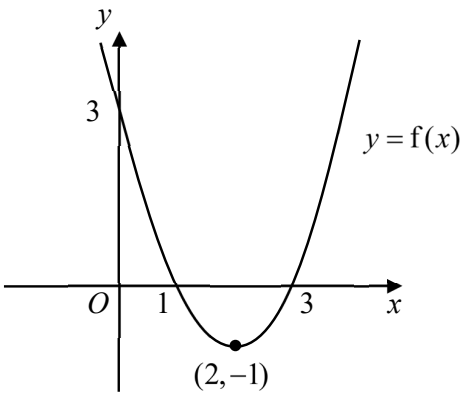


(ii)



Example 6

The diagram shows the graph of $y = \mathrm{f}(x)$.



Sketch, on separate diagrams, the graphs of

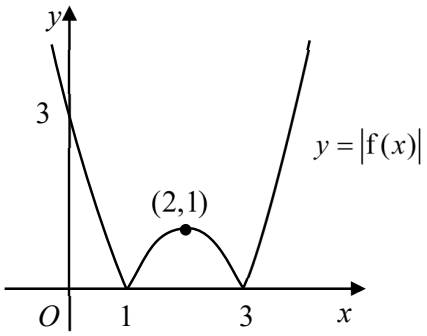
(i) $y = \left|\mathrm{f}(x)\right|$,

(ii) $y = \mathrm{f}(-x)$,

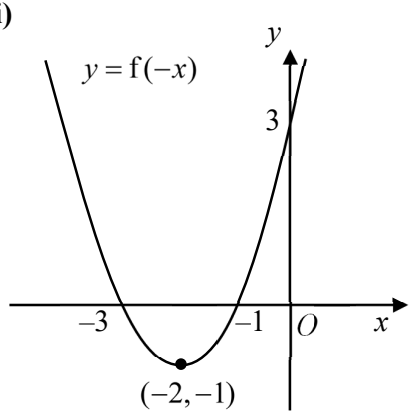
(iii) $y = \mathrm{f}\left(\left|x\right|\right)$.

Solution

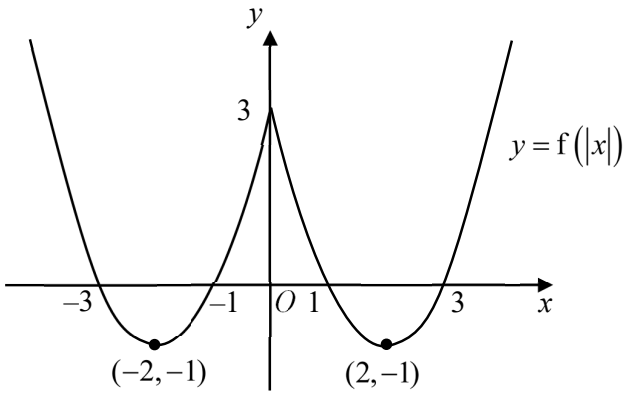
(i)



(ii)



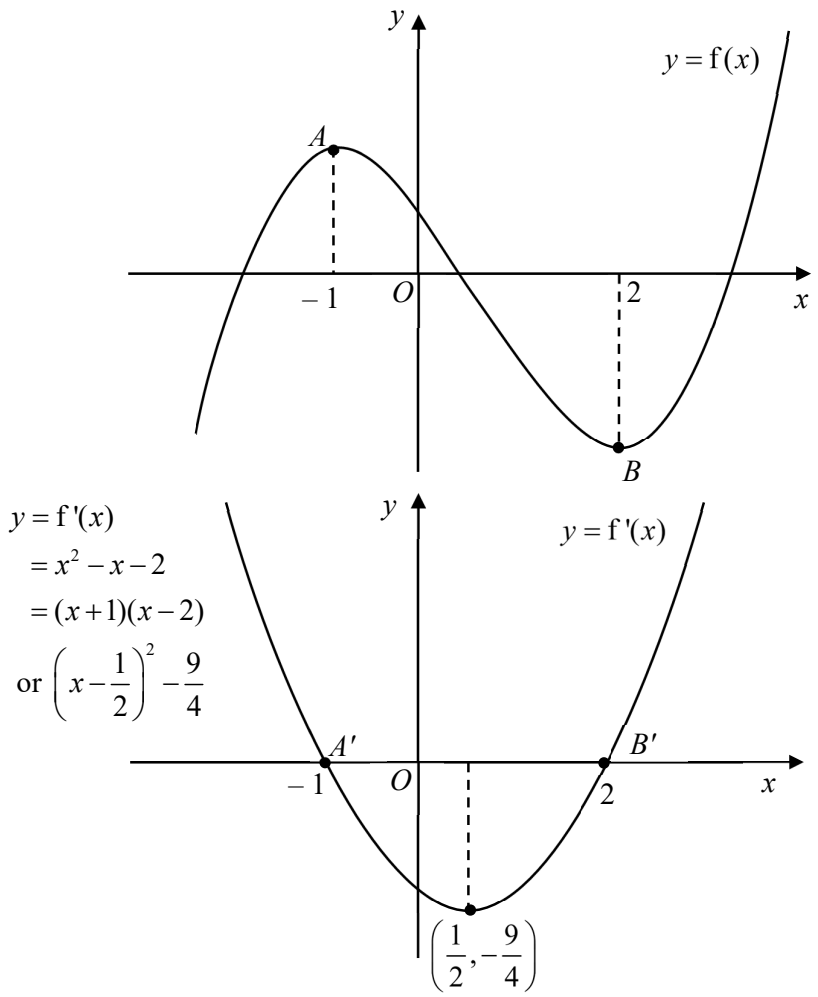
(iii)



2.2 Graphs of derivative functions

Some of the key things we want to consider when we sketch the graph of $y = \mathrm{f}'(x)$ from the graph of $y = \mathrm{f}(x)$ are the stationary points, monotonicity and concavity of the graph of $y = \mathrm{f}(x)$.

To understand the relationship between the graph of a function f and its gradient function f', let's study the example of the graphs of a cubic function, $\mathrm{f}(x) = \frac{x^3}{3} - \frac{x^2}{2} - 2x + 1$ and its derivative, $\mathrm{f}'(x) = x^2 - x - 2$.



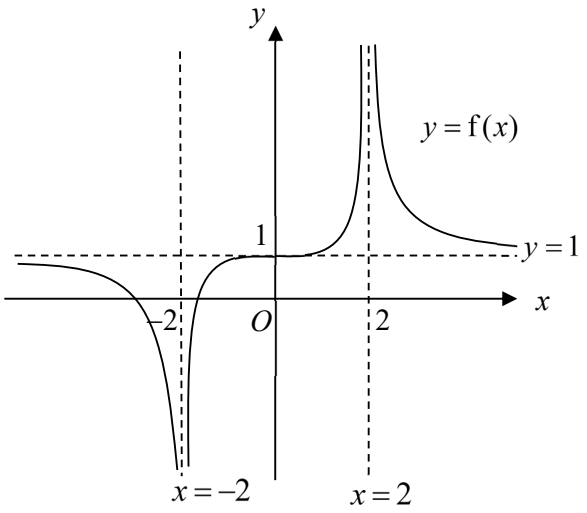
The following table provides a useful guide for sketching the graph of $y = \mathrm{f}'(x)$ from the graph of $y = \mathrm{f}(x)$.

Graph of $y = \mathrm{f}(x)$ (Note the description of the gradient)	**Graph of $y = \mathrm{f}'(x)$, that is graph of gradient of $y = \mathrm{f}(x)$**
Stationary point at $x = k$ (Gradient of stationary point is 0)	$x-$intercept at $x = k$
Strictly increasing over (a,b) (Gradient of graph is positive for (a,b))	Above $x-$axis over (a,b)
Strictly decreasing over (a,b) (Gradient of graph is negative for (a,b))	Below $x-$axis over (a,b)
Vertical asymptote $x = k$ (Gradient of vertical line is undefined)	Vertical asymptote $x = k$
Horizontal asymptote $y = k$ (Gradient of any horizontal line is 0)	Horizontal asymptote $y = 0$
Oblique asymptote $y = kx + c$ (Gradient of $y = kx + c$ is k)	Horizontal asymptote $y = k$

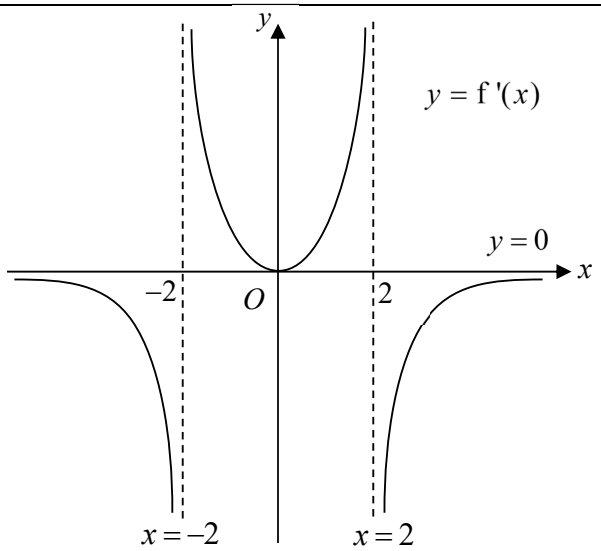
Example 7

The following diagram shows the graph of $y = \mathrm{f}(x)$. The curve has asymptotes $x = 2,\ x = -2$ and $y = 1$ and there is a stationary point of inflexion at $x = 0$.

Sketch the corresponding graph of $y = \mathrm{f}'(x)$.



Solution



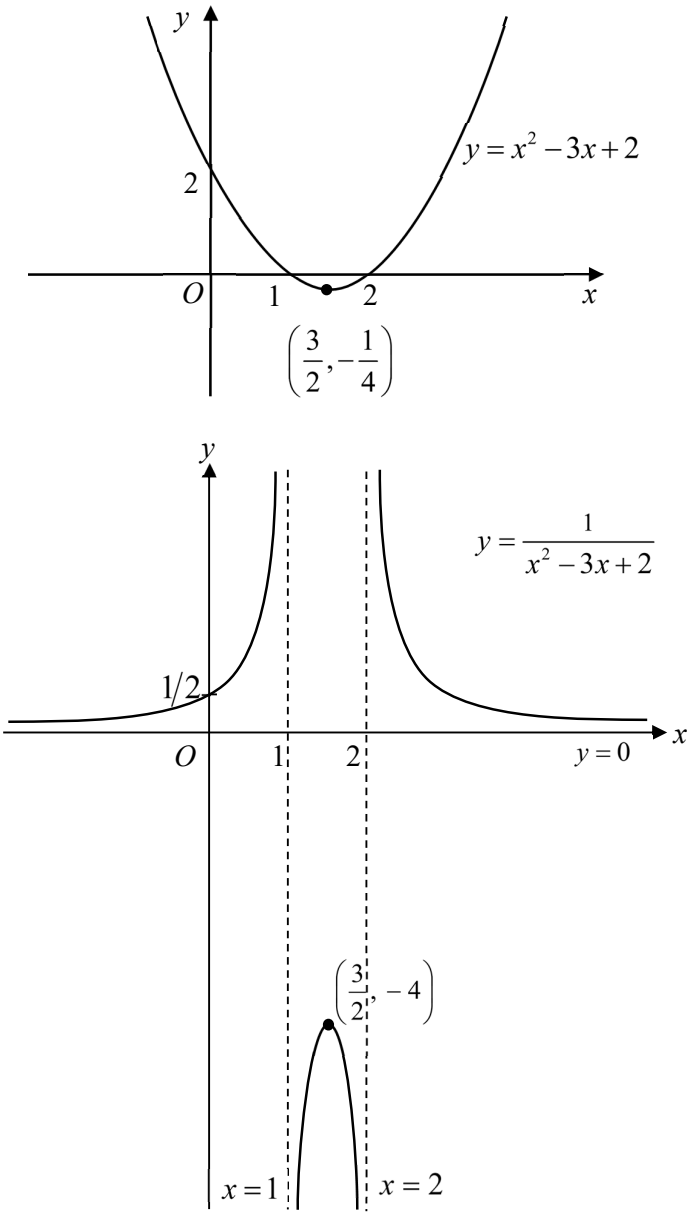
2.3 Sketching the graph of $y = \dfrac{1}{\mathrm{f}(x)}$ from the graph of $y = \mathrm{f}(x)$

It is useful to consider the following relationships between $y = \mathrm{f}(x)$ and $y = \dfrac{1}{\mathrm{f}(x)}$:

SAIL	Features of $y = \mathrm{f}(x)$	$y = \mathrm{f}(x)$	$y = \dfrac{1}{\mathrm{f}(x)}$
Intercepts and Asymptotes	x-intercepts	$(a, 0)$	Vertical asymptote $x = a$
	Asymptotes	Vertical: $x = a$	x-intercept *
		Horizontal: $y = b,\ b \neq 0$	Horizontal: $y = \dfrac{1}{b}$
		Horizontal: $y = 0$	$\dfrac{1}{\mathrm{f}(x)} \to \infty$ and/or $-\infty$
		Why ? When $\mathrm{f}(x) \to 0^+$, $\dfrac{1}{\mathrm{f}(x)} \to \infty$; when $\mathrm{f}(x) \to 0^-$, $\dfrac{1}{\mathrm{f}(x)} \to -\infty$.	
		Oblique: $y = ax + b$	Horizontal: $y = 0$
Shape	Turning points	Max pt $(h, k),\ k \neq 0$	Min pt $\left(h, \dfrac{1}{k}\right)$
		Min pt $(h, k),\ k \neq 0$	Max pt $\left(h, \dfrac{1}{k}\right)$
	y-values	(x, y)	$\left(x, \dfrac{1}{y}\right)$
		$\mathrm{f}(x) = \pm 1$	$\dfrac{1}{\mathrm{f}(x)} = \pm 1$
		Above x-axis	Remains
		Below x-axis	Remains
		$\mathrm{f}(x)$ increases	$\dfrac{1}{\mathrm{f}(x)}$ decreases
		$\mathrm{f}(x)$ decreases	$\dfrac{1}{\mathrm{f}(x)}$ increases
	Behaviour of graph	$\mathrm{f}(x) \to \infty$	$\dfrac{1}{\mathrm{f}(x)} \to 0^+$
		$\mathrm{f}(x) \to -\infty$	$\dfrac{1}{\mathrm{f}(x)} \to 0^-$

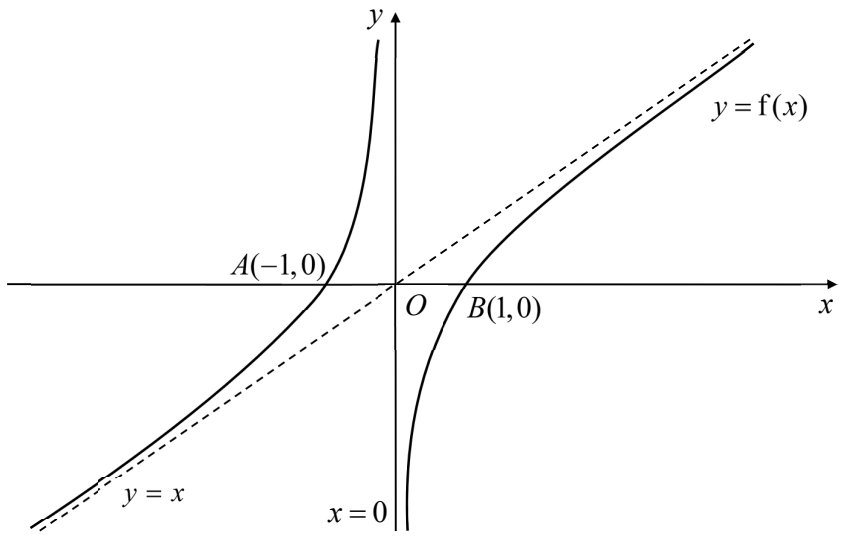
*Note : When the graph of $y = \mathrm{f}(x)$ has a vertical asymptote at $x = a$, it would imply that $\mathrm{D_f}$ does not include a. As such, the graph of $y = \dfrac{1}{\mathrm{f}(x)}$ would not include the point when $x = a$. We indicate the exclusion of the point $x = a$ with an "open" circle.

Let us look at an example, paying attention to the intercepts, stationary point(s) and behaviour of the graphs of the quadratic function, $\mathrm{f}(x) = x^2 - 3x + 2$ and its reciprocal.



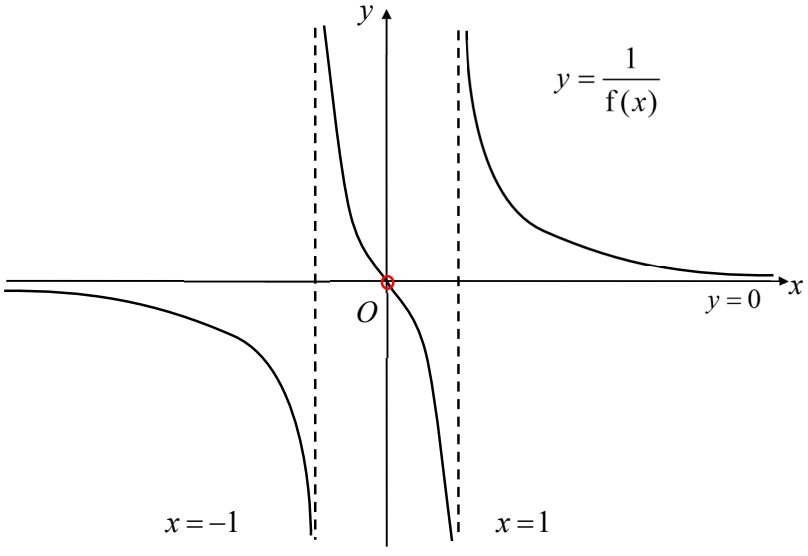
Example 8

The diagram below shows the graph of $y = \mathrm{f}(x)$. The curve has x-intercepts at A and B. The lines $x = 0$ and $y = x$ are asymptotes to the curve.



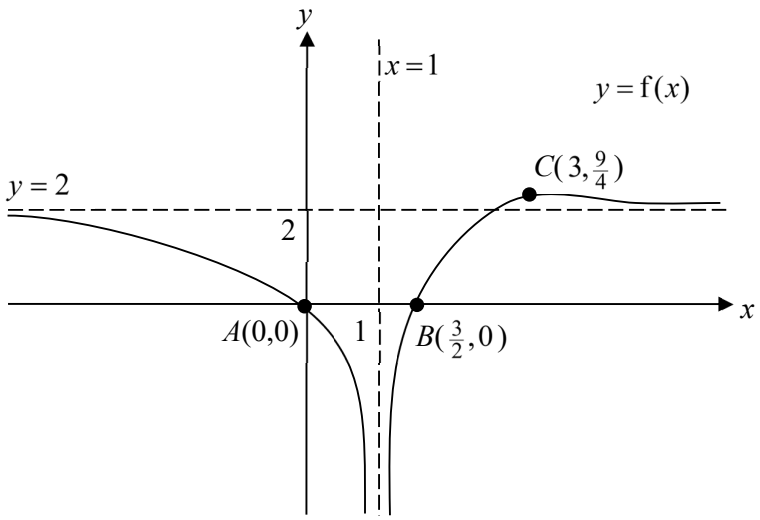
Sketch the graph of $y = \dfrac{1}{\mathrm{f}(x)}$, showing clearly the equations of asymptotes.

Solution



Example 9 RJC Y5CT 9740/2006/Q2 (modified)

The diagram below shows the graph of $y = \mathrm{f}(x)$. The curve has a maximum point at C and the lines $x = 1$ and $y = 2$ as asymptotes.



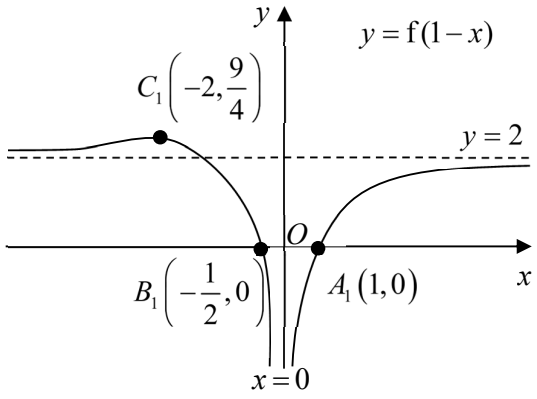
On separate diagrams sketch the graphs of

(i) $y = \mathrm{f}(1-x)$, **(ii)** $y = \dfrac{1}{\mathrm{f}(x)}$, **(iii)** $y = \mathrm{f}'(x)$,

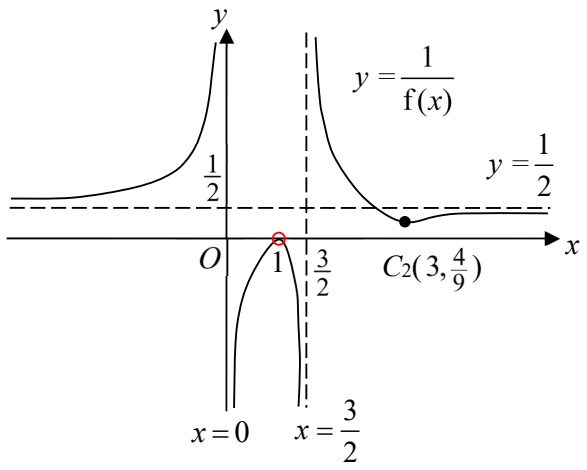
showing clearly in each case, where appropriate, the asymptotes and co-ordinates of the points corresponding to A, B and C.

Solution

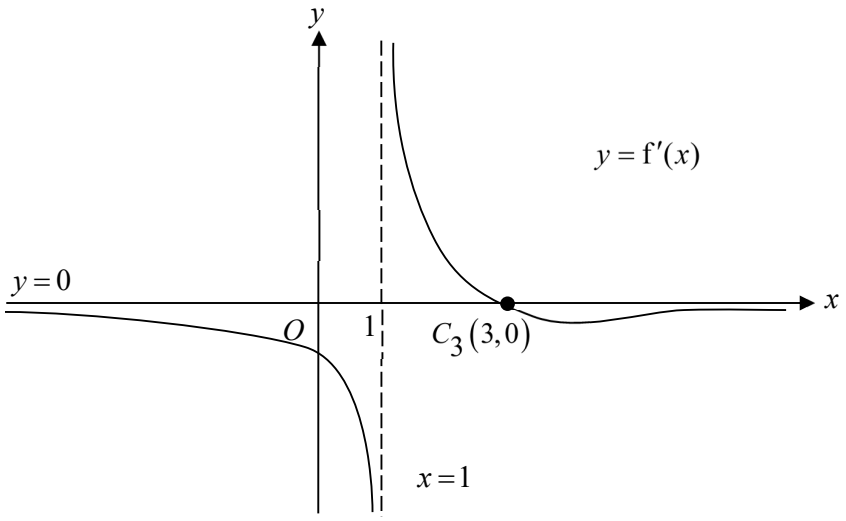
(i)



(ii)



(iii)



CONCLUSION

The exploration of transformations of graphs proves to be an illuminating journey into the dynamic nature of mathematical functions. Through translations, reflections, and scaling, we unveil the capacity to reshape and redefine the graphical representation of functions, offering a nuanced perspective on their behaviour.

Transformations of graphs stand as a testament to the elegance and universality of mathematical concepts, inviting further exploration and discovery in the ever-evolving landscape of mathematical inquiry.

SUMMARY